

Vaibhav Sharma

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linkedin.com/in/vaibhav7766 | github.com/vaibhav7766 (Featuring well-documented CV and MLOps projects)

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About Me

Data Scientist with expertise in computer vision, NLP, and MLOps. Passionate about developing scalable machine learning pipelines and innovative AI solutions.

Education

Symbiosis Institute of Technology, Pune, BTECH in AI & ML Aug 2022 - Present

- CGPA: 8.25/10 (Top 5% of batch)
- **Coursework:** Statistics, Exploratory Data Analysis(EDA) and Visualization, Data Preprocessing, Machine Learning, Deep Learning, Computer Vision, GANs, LLMs, AWS, Google Cloud, MERN Stack, Flask

Data Science & ML Projects

Image Captioning: Bridging gap for Visually challenged people github.com/vaibhav7766/CV_GAN_Project

- Developed an AI-based image captioning system using EfficientNetB4 and DeiT with a transformer decoder, optimized for edge devices, achieving a 91% F1-score on the MS COCO dataset (25GB) to generate real-time captions and audio feedback for visually impaired users.
- Tools Used: Python, PyTorch, Torchvision, Transformers, Raspberry Pi.

EasyML github.com/vaibhav7766/easymml

- Led the development of automated machine learning pipelines, reducing model deployment time by 55%. Built end-to-end ML workflows using Python, FastAPI, and Scikit-Learn to streamline operations and enhance scalability.
- Tools Used: Python, FastAPI, Java, Springboot, Scikit-Learn, TailwindCSS, PostgreSQL.

Patents

- **Offline Voice-Activated Smart Glasses System for Real-Time Visual Scene Captioning and Auditory Assistance** – Indian Patent Application No. 202521103986 A, published on Nov 21, 2025.
An offline assistive wearable system for the visually impaired.
- **System and Method for Multimodal Deepfake Detection Using Computer-Implemented Deep Learning Architecture** – Indian Patent Application No. 202521111155 A, published on Dec 19, 2025.
A robust deep learning pipeline to detect manipulated audio-visual media.
- **A Multimodal Knowledge Graph-Based System for Contextual Memory Capture and Retrieval Using Artificial Intelligence** – Indian Patent Application No. 202621044058 A, published on May 29, 2026.
An AI framework that captures and retrieves contextual memory in real-time.

Experience

Data Scientist Intern, Smart Ship Hub – Pune Jan 2026 - Present

- Worked on maritime data quality management pipelines, improving data reliability by 32% and consistency of vessel telemetry.
- Cleaned and standardized high-frequency sensor streams and noon report data, reducing preprocessing time by 45% to support downstream analytics.
- Built recommendation-driven workflows for predictive maintenance to help reduce unplanned vessel downtime by 20%.

Digital Services Operation Management Intern, BMC Software – Pune Jan 2025 - June 2025

- Transformed complex XML files into advanced XMI views and led content migration to the Helix dashboard, integrating Helix GPT for enhanced user experience using Java, PostgreSQL, and Spring Boot, boosting query performance by 30%.
- Deployed and optimized a scalable, AWS-hosted solution, ensuring 99.9% uptime and robust performance and reliability for enterprise-grade dashboard functionalities.

Full Stack Developer, Janhvi Foundation – NGO in Pune July 2024 - Dec 2024

- Developed and deployed a comprehensive community outreach platform for Janhvi Foundation using Next.js, Express, and SQL, optimizing performance on AWS for scalability and reducing page load times by 40%.
- Enhanced digital outreach by integrating welfare scheme insights into the platform, driving a 50% increase in community engagement, and improving accessibility and awareness.

Publications

Generating Descriptive Captions from Images: An AI-Based Approach for Visual Understanding May 2026

V. Sharma, S. Goyal, et al.

2026 Innovations in Machine, Engineering, and Digital Conference (IMED)

Proposes an edge-optimized transformer-based model for real-time image captioning.

DOI: 10.1109/IMED68921.2026.11484187

Skills

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| • Programming Languages: Python, Java, JavaScript, R, C, C++ | • MLOps & Tools: Jupyter, MLflow, DVC, AWS, Git |
| • Core ML/DS: pandas, NumPy, scikit-learn, TensorFlow, PyTorch, NLP, Statistics | • Data Visualization: Matplotlib, Seaborn, Tableau |
| • Web Development: MERN Stack, FastAPI, Spring Boot | • Databases: MySQL, PostgreSQL, MongoDB, NoSQL |

Certifications and Achievements

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| • Google Cloud Computing Foundations (2024) | • Google Cloud Arcade Champion (2025) |
| • Winner of GoCodeo Hackathon (2025) | • Career Essentials in GenAI with Microsoft and LinkedIn (2024) |
| • Nutanix Certified Associate (NCA) (2026) | |